

MAGNUS DIERKING

Born in Germany, 21 September 1998

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WORK EXPERIENCE

08/2025– Present Research Assistant

IAS Intelligent Autonomous Systems Group · Technical University of Darmstadt
Developing a Python/C++ software stack for FR3 robotic arms, built on ROS2 Jazzy.

10/2024– 05/2025 Research Intern

Huawei R&D Huawei Technologies Research & Development UK Limited · London, Shenzhen
Extracurricular internship during Master studies. Working on reinforcement learning, imitation learning and supporting middleware for robotics as part of the Embodied AI team.

09/2023– 09/2024 Research Assistant

IAS Intelligent Autonomous Systems Group · Technical University of Darmstadt
Assisted Ph.D. researchers with experiments, visualization and studies on supporting theory in optimal transport, Bayesian inference and optimization for robot learning.

2018–2021, 2022–2023 Working Student

mas Medical Airport Service GmbH · Frankfurt
Support personel for the rescue station on the apron of Frankfurt International Airport.

2021–2022 Teaching Assistant

TEMF Institute for Accelerator Science and Electromagnetic Fields · Technical University of Darmstadt
Developed lesson plans, theoretical / programming tutorials and provided mentorship and feedback for undergraduate students.
Courses: Electrodynamics, Numerics for Electromagnetic Field Simulation

PUBLICATIONS

2025 Ark

arXiv Open-source, Python-first robotics framework that provides a Gym-style interface for collecting data, training policies, and switching seamlessly between simulation and real-robot deployment. It includes reusable modules for control, SLAM, motion planning and visualization, and integrates natively with ROS to accelerate end-to-end robotics research.
First Author

2025 OpenPyro-A1

IEEE RA-L Open-source, low-cost bimanual half-humanoid robot designed for advanced manipulation research. It features a modular, repairable hardware design and supports coordinated two-handed tasks such as folding, cutting, and assembling. The platform enables teleoperation via a Meta Quest 3 and provides interfaces for learning-based controllers to support scalable, real-world robotics experimentation.
Co-Author

EDUCATION

<i>Master of Science</i>	10/2022– <i>Present</i>	Computational Engineering
	GPA: 1.00 · Technical University of Darmstadt Thesis: Model Tensor Planning Ongoing. Advisor: Prof. Jan PETERS · Supervision: Dr. João CARVALHO, Dr. An Thai LE Robot Learning · Reinforcement Learning · Deep Generative Models · Intelligent Robotic Manipulation · Parallel Programming Differential & Riemannian Geometry · Numerical Linear Algebra · Geometric Algebra · Information Theory 1-2 · Convex Optimization · Graph Signal Processing · Control Theory · Optimal Transport	
<i>Bachelor of Science</i>	10/2019– 10/2022	Computational Engineering
	GPA: 1.51 · Top 10% · Technical University of Darmstadt Thesis: Parallel Solution of Linear Systems Arising in Domain Decomposition Methods C++ implementation of a parallel solver for large-scale surface PDEs within an Isogeometric Analysis (IGA) framework. Advisor: Prof. Sebastian SCHÖPS · Supervision: MSc. Maximilian NOLTE · Grade: 1.00 Robotics · Software Engineering · Algorithms and Data Structures · Functional & Object-oriented Programming · Geometric Modelling Partial Differential Equations · Numerical and Statistical Methods · Mechanics 1-3 · Signals and Systems · Electrodynamics 1-2 · Maths 1-4	
<i>A-Levels</i>	10/2018– 10/2019	Mechanical Engineering
	Completed Coursework · Technical University of Kaiserslautern Material Science 1-2 · Material Science Lab · Mechanical Design Project · Production Engineering · Business Organization	
	2018	Elisabeth-Langgässer Gymnasium
	Math · English · History	

SKILLS

<i>OS</i>	LINUX · Mostly Ubuntu, some Arch
<i>Programming</i>	PYTHON · JAX, PyTorch, CVX, NumPy, SciPy, Pandas C++ · Eigen, OpenMP Basics · Java, HTML, JavaScript, CSS
<i>Simulation</i>	MuJoCo, PyBULLET
<i>Tools</i>	GIT, ROS2, Docker, L ^A T _E X
<i>Hardware</i>	Franka Research 3, Trossen Viper OptiTrack

OTHER INFORMATION

<i>Scholarships</i>	2024 · Erasmus Placements Program
	2023 · Deutschlandstipendium
<i>Languages</i>	GERMAN · Mothertongue
	ENGLISH · Fluent
	FRENCH · Basics